

In [1]: !nvidia-smi

```
Wed Jun 3 04:16:39 2026
+-----+
+-----+
| NVIDIA-SMI 570.172.08           Driver Version: 570.172.08      CUDA Version: 12.
8      |
|-----+-----+-----+-----+
+-----+
| GPU  Name                Persistence-M | Bus-Id                Disp.A | Volatile Uncor
r. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |      Memory-Usage | GPU-Util  Compu
te M.  |
|      |      |      |      |      |      |      |      |      |      |      |      |
|      |      |      |      |      |      |      |      |      |      |      |      |
IG M.  |
|=====+=====+=====+=====+
=====|
|  0  NVIDIA H200                On  |  00000000:61:00.0 Off |
On  |
| N/A   30C    P0                75W / 700W |      N/A      |      N/A      De
fault  |
|      |      |      |      |      |      |      |      |      |      |      |      |
abled  |
+-----+-----+-----+-----+
+-----+

+-----+
+-----+
| MIG devices:
|
+-----+-----+-----+-----+
+-----+
| GPU  GI   CI   MIG |      Memory-Usage |      Vol |      Shared
|      |      |      |      |      |      |      |      |      |      |
|      |      |      |      |      |      |      |      |      |      |
|      |      |      |      |      |      |      |      |      |      |
JPG  |      |      |      |      |      |      |      |      |      |
|      |      |      |      |      |      |      |      |      |      |
|      |      |      |      |      |      |      |      |      |      |
|=====+=====+=====+=====+
=====|
|  0   9   0   0 |      15MiB / 16384MiB |  16      0 |  1   0   1   0
1  |
|      |      |      |      |      |      |      |      |      |      |
|      |      |      |      |      |      |      |      |      |      |
|      |      |      |      |      |      |      |      |      |      |
+-----+-----+-----+-----+
+-----+

+-----+
+-----+
| Processes:
|
| GPU  GI   CI           PID   Type   Process name                      GPU M
emory |
|      |      |      |      |      |      |      |      |      |      |
|      |      |      |      |      |      |      |      |      |      |
|      |      |      |      |      |      |      |      |      |      |
|=====+=====+=====+=====+
=====|
| No running processes found
```

|

+-----+

-----+

```
In [7]: import torch
print("pytorch version :",torch.__version__)
print("cuda available:",torch.cuda.is_available())
print("Gpu count:",torch.cuda.device_count())
if torch.cuda.is_available():
    print("Gpu name:",torch.cuda.get_device_name(0))
```

```
pytorch version : 2.5.0a0+e000cf0ad9.nv24.10
cuda available: True
Gpu count: 1
Gpu name: NVIDIA H200 MIG 1g.18gb
```

```
In [8]: device = torch.device("cuda" if torch.cuda.is_available() else "cpu" )
print(device)
```

```
cuda
```

```
In [9]: x=torch.tensor([1,2,3])
print(x)
print(x.device)
```

```
tensor([1, 2, 3])
cpu
```

```
In [10]: x_gpu=x.to(device)
print(x_gpu)
print(x_gpu.device)
```

```
tensor([1, 2, 3], device='cuda:0')
cuda:0
```

```
In [11]: import time
A=torch.rand(5000,5000).to(device)
B=torch.rand(5000,5000).to(device)
start=time.time()
c=torch.matmul(A,B)
end=time.time()
print("Execution time:",end-start)
```

```
Execution time: 0.07146096229553223
```

```
In [18]: x=torch.tensor([
    [8.5,80,90],
    [6.0,45,40],
    [9.2,95,92],
    [4.8,25,30],
    [7.8,85,80],
    [5.0,35,38]
],dtype=torch.float32).to(device)
y=torch.tensor([
    [1],
    [0],
    [1],
    [0],
```

```

        [1],
        [0]
    ], dtype=torch.float32).to(device)
    import torch.nn as nn
    model=nn.Sequential(
        nn.Linear(3,16),
        nn.ReLU(),
        nn.Linear(16,8),
        nn.ReLU(),
        nn.Linear(8,1),
        nn.Sigmoid()
    ).to(device)
    print(model)
    loss_fn=nn.BCELoss()
    optimizer=torch.optim.Adam(
        model.parameters(),
        lr=0.01
    )

```

```

Sequential(
  (0): Linear(in_features=3, out_features=16, bias=True)
  (1): ReLU()
  (2): Linear(in_features=16, out_features=8, bias=True)
  (3): ReLU()
  (4): Linear(in_features=8, out_features=1, bias=True)
  (5): Sigmoid()
)

```

```

In [22]: device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
         print("Device:", device)

```

Device: cuda

In []:

In []:

In []:

In []: